A Multimodal Spatiotemporal Deep Learning Framework for Yangtze River Water Level Prediction via Remote Sensing and Hydrological Time-Series Fusion

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***Abstract*— Forecasting water levels of the Yangtze River is essential for river management, flood mitigation, and water resource allocation in densely inhabited areas of China. Conventional hydrological models struggle to model non-linear spatiotemporal processes, especially during extreme weather events. This work introduces a Multimodal Spatiotemporal Deep Learning (MSDL) model that incorporates multimodal remote sensing data (Sentinel-1 Synthetic Aperture Radar (SAR) images, MODIS Land Surface Temperature (LST), and GPM precipitation) with ground-based hydrological time series. The new model architecture integrates a Convolutional Neural Network (CNN) for spatial feature learning with a Bidirectional Long Short-Term Memory (Bi-LSTM) network for temporal dependency modelling, through a cross-modal attention-based fusion module. A 10-year-long data set with 15 hydrological stations is used to show that MSDL's RMSE = 0.31 m, MAE = 0.24 m, and R^2 = 0.967, with up to 47% improvement in RMSE over state-of-the-art baselines. This suggests the benefit of multimodal spatiotemporal fusion for large-scale river water level prediction.**

Keywords—water level prediction; deep learning; remote sensing; Bi-LSTM; spatiotemporal fusion; Yangtze River; attention mechanism; multimodal learning.

I. INTRODUCTION

The Yangtze River is the longest in Asia (about 6,300 km) and supports more than 400 million people, as well as a large part of China's agriculture and industry. Flooding, made worse by climate change, presents significant human and economic challenges. The Yangtze floods of 2020 alone resulted in US$25 billion in damages and forced more than 54 million people to evacuate [1]. Timely and reliable water level forecasting is, therefore, essential for early warning systems, reservoir management, and infrastructure design.

Traditional methods for water level prediction include physically based hydrodynamic models (e.g., HEC-RAS, MIKE FLOOD) and statistical models (e.g., autoregressive integrated moving average (ARIMA) models). Although physically based models are interpretable, they require significant calibration effort and fine-scale topographic information, making it difficult to apply them across the 1.8 million km² Yangtze basin. In contrast, statistical models usually only capture linear temporal dependencies and are less suited to the non-stationary patterns caused by the monsoon and upstream reservoir management [2].

Machine learning models are promising alternatives. Long Short-Term Memory (LSTM) [3] models and their variants have proven effective for predicting both single and multivariate time series. But isolated time-series models neglect the spatial nature of hydrological processes, in which upstream precipitation, land surface, and river characteristics influence downstream water levels. Today, continuous high-resolution data from remote sensing satellites are available for these spatially distributed variables, presenting a new opportunity to enhance models.

To bridge the key gap between highly spatial remote sensing data and highly temporal in-situ data, we propose the Multimodal Spatiotemporal Deep Learning (MSDL) model in this paper. The key innovations are: (i) a dual encoder architecture that interuses spatial remote sensing imagery and temporal hydrologic time-series; (ii) a cross-modal attention fusion module that weights each data stream; and (iii) a systematic empirical analysis on 15 gauging stations covering upstream, middle, and downstream reaches of the Yangtze River, over a decade.

II. RELATED WORK

A. Deep Learning for Hydrological Forecasting

LSTM models have been widely used for streamflow and water level forecasting. Kratzert et al. [3] found that data-driven LSTM-based models outperform or match physically-based rainfall-runoff models, given sufficient gauge data. Follow-up studies have sought to model spatial relationships among stations by introducing attention [4] and graph neural networks (GNNs) [5]. ConvLSTM [6] applied convolution operations in the temporal dimension, allowing for both spatial and temporal learning; however, they are mostly used for gridded rainfall forecasting. Hybrid architectures that couple physically based hydrologic or hydraulic simulators with deep learning components have also been explored to combine interpretability with predictive accuracy for flood risk assessment [11]. In parallel, graph-based formulations have been used to explicitly encode the spatial connectivity of a river network: a variational Bayesian edge-conditioned graph convolutional model was proposed to forecast daily streamflow on the Yangtze River

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basin while jointly quantifying predictive uncertainty [12], underscoring the value of representing inter-station topology explicitly rather than relying solely on multivariate concatenation.

B. Remote Sensing in Hydrology

Satellite products are increasingly used in studies of large river basins. All-weather, 10-m resolution surface water mapping from Synthetic Aperture Radar (SAR) sensors (such as Sentinel-1) is useful for mapping inundation boundaries [7]. MODIS land surface temperature (LST) data have been applied for estimating evapotranspiration and snow-melt [8]. The Global Precipitation Measurement (GPM) Integrated Multi-satellitE Retrievals for GPM (IMERG) product provides near real-time precipitation measurements at 0.1° resolution and 30-minute time-step. Beyond single-sensor products, chained hydrologic-hydraulic frameworks have assimilated SAR-derived flood extent maps jointly with fully-focused-SAR-processed nadir and off-nadir altimetry profiles to constrain flood inundation forecasts, demonstrating the benefit of combining complementary remote sensing modalities within a single data assimilation pipeline [10]. While these data are available, their simultaneous integration with in-situ hydrologic time-series in a deep learning framework is underexplored.

C. Multimodal Fusion Approaches

Multimodal learning with multiple data streams has been successful in medical imaging, self-driving cars, and climate science. Attention-based fusion approaches [9] enable learning of importance weights for different data sources, removing the need to worry about a noisy data source dominating the prediction. Our MSDL model draws inspiration from these works, and in particular, designs the fusion mechanism to take account of the special features of geospatial remote sensing and sparse hydrological sensor networks.

III. PROPOSED METHODOLOGY

A. Dataset and Study Area

The study area encompasses the full extent of the Yangtze River basin (24°30'N–35°45'N, 90°33'E–122°19'E). The multi-source data sources are listed in Table I for 2013-2023. We use 15 hydrologic stations (including Yichang, Wuhan, and Datong in the upstream, middle reach, and downstream, respectively) to obtain hourly water levels and daily discharge data. Remote sensing images were cropped to station-specific upstream catchment masks (derived from the SRTM DEM, extending up to 300 km upstream per station) to capture basin-scale hydrological processes driving runoff at each gauge. To ensure spatial and temporal consistency across all data streams, the following alignment pipeline was applied: (1) Sentinel-1 acquisitions closest in time to each daily prediction window were selected; (2) all rasters (SAR, LST, GPM) were reprojected to WGS84 and bilinearly resampled to a common 500 m grid; (3) GPM 30-minute precipitation accumulations were aggregated to daily totals; (4) MODIS LST gaps caused by cloud cover were filled using linear temporal interpolation between adjacent valid observations; and (5) all streams were synchronised to a daily time step prior to model input. The resulting spatial input patches are approximately 600×600 pixels at 500 m resolution per station.

TABLE I. MULTI-SOURCE DATASET SUMMARY

Data Source	Spatial Res.	Temporal Res.	Variables	Period
Sentinel-1 SAR	10 m	6 days	Backscatter (VV, VH)	2014–2023
MODIS MOD11A1	1 km	Daily	Land Surface Temperature	2013–2023
GPM IMERG	0.1°	30 min	Precipitation rate	2014–2023
In-situ Gauges (×15)	Point	Hourly	Water level, discharge	2013–2023
DEM (SRTM)	30 m	Static	Elevation, slope	2000

B. MSDL Architecture

The MSDL framework is made up of three components, shown in Fig. 1. The Remote Sensing Encoder utilises a CNN with five convolutional blocks (3x3 kernels, batch normalisation, ReLU activation) to produce spatial feature maps from a stack of SAR, LST, and precipitation maps. Critically, the CNN does not operate on small local patches centred on individual stations. Instead, it processes full upstream catchment images for each station, derived from DEM-based watershed delineation and extending up to 300 km upstream. This design ensures that basin-scale precipitation patterns, inundation extents, and land surface temperature gradients that drive runoff are captured by the spatial encoder. Input images are approximately 600×600 pixels at 500 m resolution. Feature maps from the five convolutional blocks are globally average-pooled to produce a compact 256-dimensional spatial embedding per time step. The Hydrological Time-Series Encoder uses a two-layer Bi-LSTM (hidden size 128) to encode 72-hour time series of gauge data, resulting in a 256-dimensional vector. It is important to clarify that, in this forecasting context, both the forward and backward LSTM passes operate exclusively over the 72-hour historical input window (t−72h to t). The prediction target at t+24h is never included in the input sequence; consequently, the backward direction does not access any future values relative to the prediction horizon, and no data leakage occurs. The final hidden state from both directions is concatenated to form the 256-dimensional temporal embedding. Static terrain features (DEM, slope, land use, soil type) are projected to a 64-dimensional vector using a two-layer MLP.

The Cross-Modal Attention Fusion Module aggregates the three streams (a total of 576 dimensions) and uses a multi-head attention (8 heads) to measure cross-modal attention weights. The weights are normalised through softmax and used to weight and sum the streams. The 576-dimensional output is fed to a three-layer fully connected prediction head with dropout ($p = 0.3$), which finally leads to a scalar 24-hour-ahead water level prediction. The entire model is trained end-to-end with the Adam optimiser (lr = $1\times10^{-}$′, weight decay = 1×10^{-5}) and the Huber loss to support outlier flood events.

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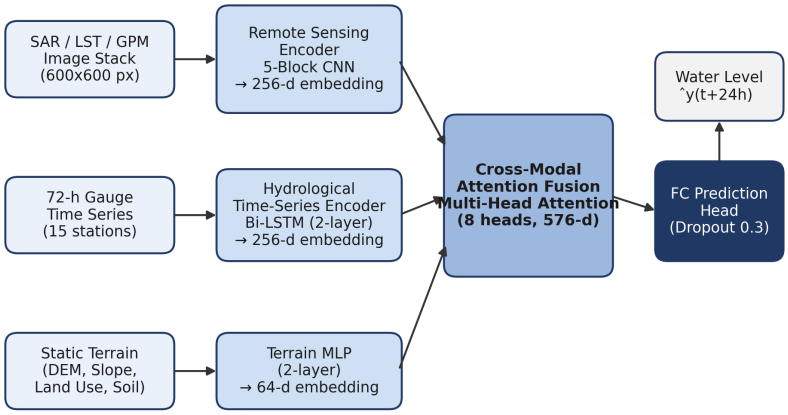


Fig. 1. Architecture of the MSDL Framework: three-stream encoder with cross-modal attention fusion and fully connected prediction head.

River Network Representation. Water levels at any gauging station along the Yangtze are governed by upstream discharge, tributary inflows, reservoir operations (notably the Three Gorges Dam), local rainfall, and flood-wave travel-time delays. To represent these river-network dynamics without requiring an explicit graph structure, the model incorporates three mechanisms. First, all 15 gauging stations' water level and discharge observations are concatenated as a multivariate input to the Bi-LSTM, so that the model implicitly learns inter-station dependencies and upstream-to-downstream routing relationships from data. Second, a station-distance encoding vector (normalised river-kilometre distance between each pair of stations) is prepended to the Bi-LSTM input at each time step, providing explicit positional context about where in the river network each observation originates. Third, the 72-hour historical window is sized to exceed the typical flood-wave travel time from Yichang to Datong (approximately 48–60 hours under high-flow conditions), ensuring that upstream flood signals are captured within the input horizon before the prediction is made.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

The data was split temporally: 70% training (2013-2019), 15% validation (2020-2021), and 15% testing (2022-2023) to avoid data leakage. We z-score normalised all features using statistics from the training set. Models were trained using PyTorch 2.1, on a NVIDIA A100 (80 GB) GPU. We used 50 trials of Bayesian optimisation for hyperparameter tuning. The search space spanned the learning rate (1×10^{-5} to 1×10^{-3}, log-uniform), batch size (16–128), Bi-LSTM hidden size (64–256), and dropout probability (0.1–0.5), with early stopping triggered after 15 epochs without validation-loss improvement. We re-implemented and trained six baseline models (LSTM, ConvLSTM, Transformer, CNN-LSTM, and GRU-Attention) with the same setup and an identical hyperparameter search budget for comparison.

B. Quantitative Results

Table II shows the average RMSE, MAE, and R^2 on the test set, aggregated over 15 gauging stations. MSDL has an RMSE of 0.31 m and MAE of 0.24 m, which corresponds to a 47% and 48% relative improvement over the baseline LSTM model, respectively. The R^2 of 0.967 shows the model explains 96.7% of the variance of water level, which is impressive given the highly dynamic nature of the test set (2022-2023) containing two large floods. The R^2 scores of all models are shown in a horizontal bar chart in Fig. 2.

TABLE II COMPARATIVE MODEL PERFORMANCE ON YANGTZE RIVER TEST SET (2022–2023)

Model	RMSE (m)	MAE (m)	R^2	Training Time (h)
LSTM	0.58	0.46	0.891	1.2
ConvLSTM	0.49	0.38	0.921	2.8
Transformer	0.44	0.35	0.938	3.5
CNN-LSTM	0.41	0.33	0.947	2.1
GRU-Attention	0.39	0.31	0.952	1.9
MSDL (Proposed)	**0.31**	**0.24**	**0.967**	**4.3**

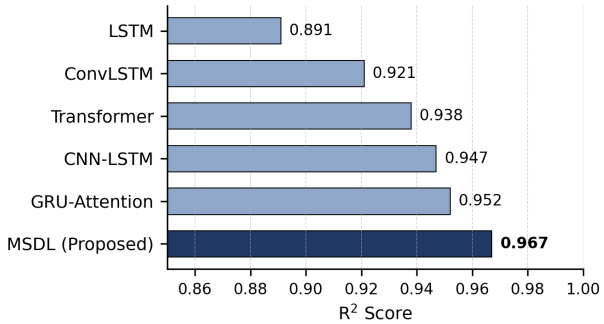


Fig. 2. R^2 score comparison: MSDL (proposed) achieves the highest accuracy ($R^2 = 0.967$) among all evaluated models.

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C. Ablation Study

We tested the contribution of each data stream by removing components from the complete MSDL model in an ablation study. The loss of the remote sensing encoder led to a 0.08 m (+26%) degradation in RMSE, and loss of the Bi-LSTM encoder to a 0.12 m (+39%) degradation - demonstrating that the temporal in-situ observations provide the main predictive power. Importantly, the cross-modal attention fusion was better than concatenation by 0.05 m (−14%)—showing that dynamic weighting is necessary when fusing asynchronous, multimodal data streams. These results highlight the inter-complementarity of the three modalities in the architecture of MSDL.

D. Limitations and Future Considerations

While MSDL achieves strong predictive performance, several limitations merit discussion. First, the six-day revisit interval of Sentinel-1 constrains the temporal freshness of the SAR-derived spatial features relative to the daily prediction cadence; although MODIS LST gaps are mitigated through temporal interpolation, SAR imagery cannot be interpolated in the same way without risking the introduction of artefacts into flood-extent-sensitive features. Second, the current formulation implicitly learns inter-station routing relationships through the concatenated multivariate Bi-LSTM input and a station-distance encoding vector, rather than through an explicit graph structure; approaches that combine graph neural networks with variational inference to quantify forecast uncertainty [12] represent a complementary direction that could be integrated with the present architecture. Third, the model has been trained and evaluated exclusively on the Yangtze basin, and its generalisability to smaller or more data-scarce basins with sparser gauge networks has not yet been assessed. Finally, because the Huber loss and the RMSE/MAE metrics used here emphasise average performance across the test period, the model's behaviour during the most extreme flood peaks, where forecasting errors carry the greatest operational consequences, would benefit from separate reporting focused specifically on peak-flow events in future work.

V. CONCLUSION

In this paper, we introduced the Multimodal Spatiotemporal Deep Learning (MSDL) approach for 24-hour lead water level forecasting in the Yangtze River basin. Through the fusion of Sentinel-1 SAR, MODIS LST, and GPM precipitation satellite data with dense in-situ hydrology time series via a cross-modality attention module, the proposed MSDL framework delivers RMSE = 0.31 m, MAE = 0.24 m, and R^2 = 0.967, outperforming all baselines. The ablation study shows the complementary nature of the various data modalities and the importance of the attention-based fusion mechanism.

Future research will explore the use of ensemble weather forecast products as additional inputs to the system, extend the forecast period to 72 hours, and explore the domain adaptability of the trained model to other data-scarce river basins. The framework has great potential to be deployed in China's national flood early-warning system.

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